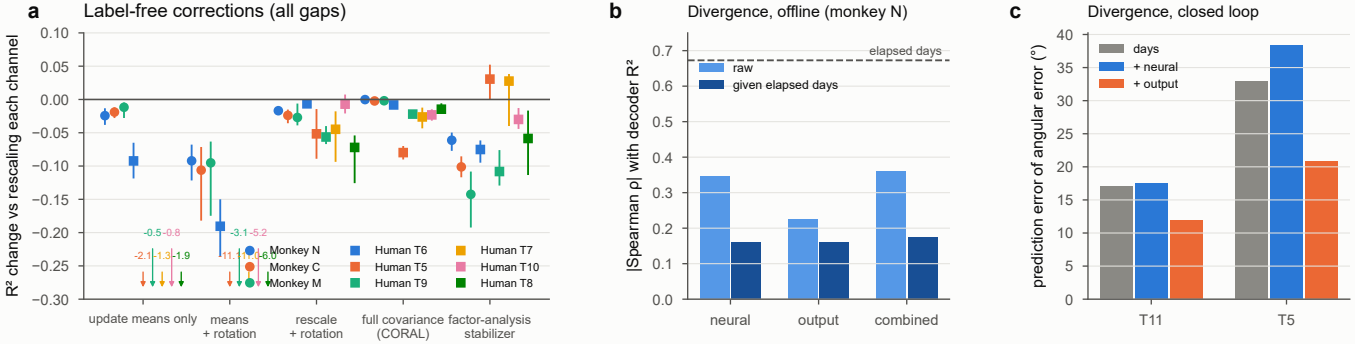

a
Label-free corrections (all gaps)
R² change vs rescaling each channel
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
-0.5
-0.8
-2.1
-1.3
-1.9
-3.1
-5.2
-11.1
-6.0
Monkey N
Monkey C
Monkey M
Human T6
Human T5
Human T9
Human T7
Human T10
Human T8
update means only
means + rotation
rescale + rotation
full covariance (CORAL)
factor-analysis stabilizer
b
Divergence, offline (monkey N)
|Spearman ρ| with decoder R²
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
elapsed days
raw
given elapsed days
neural
output
combined
c
Divergence, closed loop
prediction error of angular error (°)
40
35
30
25
20
15
10
5
0
days
+ neural
+ output
T11
T5